Thornado Cash

Nodes
- 67% FROST vault with state machine
- Nodes are New, Selected, Active or Standby
- New: New node, may be incomplete or missing preflight
- Selected: selected to churn in
- Active: consensus, observation, witnessing, keygen-keysign
- Standby: churned out, still can keysign on retired vaults, not in consensus or witnessing
- Nodes penalized for missing observations or TSS Signing
- 3-day churning, always, at least one node in standby
- Node churned out is one with most penalty points
- Network churns when selected node participates in a new keygen and migrates funds
- Every churn:
  - Baddest node
  - AND Node wish to LEAVE (if any)

Auction Slots
- Permanent node slots, once bonded, a node owns a slot
- Nodes bond 0 + 1n BTC (parameters)
- Bond can never leave the system, unless they all vote Shutdown
- Nodes can sell node slots in an auction market
- Slot is traded once highest bidder chosen
- Node must churn out, new node churns into that slot
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Shutdown
- 67% of nodes vote to shutdown
- 30 days of maintenance mode, no further deposits
- All principle returned to nodes (their original deposit)
- Fees, Abandoned deposits split equally between all nodes

ZK
- The deposit address and amount is known.
- The amount is converted to equal-denom notes
- The user proves they deposited an amount, which entitles them to a set of notes
- The user later brings a note, but the note is completely fungible with all other equal denomination notes
- The user places a withdrawal address and submits
- If the proof is valid, the withdrawal submission is gasless (no relayer)

Protection
- Privacy is attained by distance between deposit and withdrawal, controlled by the user
- If user immediately withdraws, no privacy
- Nodes cannot unilaterally move funds, no custodian service
- Nodes are regularly churned, means anyone can leave or reenter
- Permissionless entry-exit
- Funds are not churned for privacy, funds are only churned for the TSS keyset change
- Dev Fund is 10% of Node Slot sales
- Fees are charged only on outbound fees, not against transaction size

Node Income
- Charged as TSS gas handling outbound fees min 100k Sats or 1% (whichever is higher)
- Fees accumulate in bucker
- Operators can sign for their fees, which become further note deposits

Network Stack
- Landing Page links out to mirrors, X
- Client-mirrors + light nodes (one click deploy), GitHub registry release hash
- Node + BitcoinD + TSS

Client Mirrors
- All nodes must “vouch” for a client-mirror with SSL, they can run it themselves or get a third party to run it
- Client site operates a site that content hashes to a known hash - non-scam site
- If active nodes can validate the site, then nodes can enter the set
- Sites are checked once a week, if down, nodes earn penalty points
- Client also attached to light node, light node broadcasts tx and queries state machine

UX
- Users go to client site, a HD mnemonic is generated
- Users conduct a proof of work to request a deposit address (targeting 30sec on iPhone/browser)
- Users can then deposit any amount of bitcoin
- Once deposit received and confirmed, client can then convert the deposit into a set of notes with a zero knowledge proof, smallest to largest
- 10, 1, 0.1, 0.01 etc
- Each index is an index on the HD wallet path (BIP84)
- ZK Proof is proof that the user is entitled to a withdrawal, but the deposit is unowned.

Request -> Deposit -> Split -> Wait -> Withdraw